Complete spectrum of h-fold integer sumset cardinalities for fixed h and sufficiently large set size (Original Theorem 1.4)

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Editorial note. The fixed v3 arXiv abstract comments say 17 pages, but the actual publisher-formatted PDF has 24 physical pages and its first-page header says Discrete Analysis 2026:2, 24 pp. The actual PDF is used for page locators.

Editorial note. The v3 native-source grant and publisher page are CC BY 4.0; the original PDF copyright link points to CC BY 3.0. These distinct version/component grants are retained separately.

Editorial note. The original gap-exclusion proof at native line 250 prints d >= k-ell and its displayed comparison also uses k-ell beside the m+ell h expression. These exact author signs and expressions are preserved without replacement or an inferred mathematical correction.

Editorial note. The original Section 4.3.1 overview at native line 460 prints f(x_1,...,x_d)=hA and says hA is nonincreasing, whereas its neighboring definitions concern the integer sumset cardinality. The author text is retained literally.

Editorial note. The original equation (4.13) at native lines 634–636 prints the B_{j+1} minus B_j order. It is retained literally beside the original Lemma 4.13(a) difference order and the complete subsequent comparison text; no sign repair is made.

Editorial note. The original Lemma 4.13(d) proof at native line 540 writes B_{s,ell-1}=[0,ell-2], with its exact ell indexing unchanged throughout. No model-authored replacement is added.

Editorial note. Remark 1.8 explicitly reports the later fully ChatGPT-written polynomial N(h,k) proof in the separate GR26 blog. Its disclosed attribution is retained as context. No generated proof or G/H replacement construction is imported or counted.

Editorial scope. One original author Theorem1.4, with full current main proof and required h=3/gap-exclusion branches. Source attribution and faithful collection do not independently certify the mathematics. Later fully model-generated polynomial diameter result is disclosed but excluded.

Allowed standard prerequisites (editorial).

- General h-fold sumset cardinality bounds and the h=2 cardinality interval. Original equation (1.1), Nathanson [Nat96] Theorem 1.3, and Erdős–Szemerédi [ES83], retained in the original introduction and bibliography. Bounds and the standard h=2 branch of the main combination proof; all h>=3 current author constructions remain in the material
- Lev multiple-addition diameter inequality. Original Theorem 2.2 with exact equations (2.1)/(2.2), citing Lev [Lev96], Structure theorem for multiple addition and the Frobenius problem, J. Number Theory 58(1):79–88 (1996). Classical prerequisite for the current gap-exclusion proof and dense component counts; its exact statement is present, its standard proof is not recursively reproduced
- Binomial theorem, finite-dimensional Freiman-map terminology and elementary generating-function coefficient convolution. Original §§3–4 and references [Nat96]/[Lev96]; current native Lemmas 4.6, 4.9 and 4.11 themselves are fully retained. Algebraic prerequisites do not replace the current S/T normal-form counts, parameter decrement estimates or interval overlap construction

Source reference. Original section 5.2: metric-decompositions/DA2026-165102/sources/arxiv-v3.tex, source lines 757–762. Introduction-only discussion of the excluded k<=h future regime; not invoked in the selected Theorem 1.4 proof or its current core.

Source reference. Original section 5: metric-decompositions/DA2026-165102/sources/arxiv-v3.tex, source lines 656–656. Roadmap-only pointer to future questions; selected proof ends at original line 654.

1 Introduction

Let $A \subset \mathbb{Z}$ be a set of integers. For $h \in \mathbb{N}$, write

$$hA \coloneqq \{a_1 + \cdots + a_h : a_i \in A\}$$

to denote the h-fold sumset of A. Note that the a_i need not be distinct. Nathanson defined the set $\mathcal{R}(h,k)$ of the possible sizes of these h-fold sumsets when $|A| = k$.

Definition 1.1 ([Nat25c])**.** Let $h, k \in \mathbb{N}$. Define the *range of cardinalities of h-fold sumsets*

$$\mathcal{R}(h,k) \coloneqq \{|hA| : A \subset \mathbb{Z} \text{ and } |A| = k\}.$$

Nathanson [Nat25c] posed the problem of determining the exact values in $\mathcal{R}(h,k)$ for arbitrary $h, k \in \mathbb{N}$. In this paper, we will determine the form of $\mathcal{R}(h,k)$ for fixed $h \in \mathbb{N}$ and large k, and for $h = 3$ and all $k \in \mathbb{N}$, specifically answering Nathanson's Problems 8 and 9 in [Nat25c].

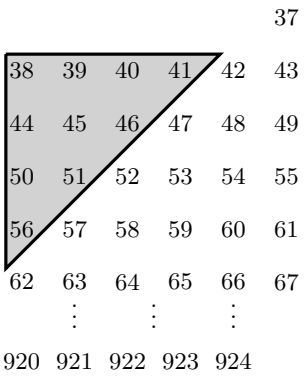


Figure 1: We trivially have $\mathcal{R}(6,7) \subseteq [37, 924]$. The gray triangle represents the numbers in $\Delta_{6,7}$ and Theorem 1.3 shows that there are no elements of $\mathcal{R}(6,7)$ in $\Delta_{6,7}$. Conjecture 1.6 says that $\mathcal{R}(6,7)$ is all of the listed numbers outside the gray triangle.

The first work related to $\mathcal{R}(h,k)$ was by Erdős and Szemerédi [ES83], who remarked that "it is easy to see that" $\mathcal{R}(2,k) = \left[2k-1, \binom{k+1}{2}\right]$ for all $k \in \mathbb{N}$. Nathanson [Nat25c] determined the exact form of $\mathcal{R}(h,3)$ for any $h \in \mathbb{N}$. Nathanson's recent program has involved finding specific values in $\mathcal{R}(h,k)$ and studying the frequency with which certain sumset sizes show up, with particular attention to $k = 4$ [Nat25a, Nat25b, Nat25c, Nat26a, Nat26b, O'B25]. With fixed $h > 2$ and k increasing, the problem of determining $\mathcal{R}(h,k)$ remained fully open.

For real numbers a and b, let $[a,b]$ denote the interval of integers from $\lceil a \rceil$ to $\lfloor b \rfloor$, including endpoints. It is not hard to show [Nat96, Theorem 1.3] that

$$\mathcal{R}(h,k) \subseteq \left[hk - h + 1, \binom{h+k-1}{h}\right], \tag{1.1}$$

and that the endpoints of this interval are achieved by the arithmetic progression $A = [0, k-1]$ and the geometric series $A = \{1, h+1, (h+1)^2, \ldots, (h+1)^{k-1}\}$. Let $m = hk - h + 1$. Tang–Xing [TX21] and Schinina [Sch26] independently found that there are no elements of $\mathcal{R}(h,k)$ in $[m+1, m+h-2]$. Tang–Xing also found that there are no elements of $\mathcal{R}(h,k)$ in $[m+h+1, m+2h-3] \cup [m+2h+1, m+3h-4]$ if $k \geq 5$. We extend this trio of intervals into a set of $\min\{h,k\} - 2$ intervals, which we call $\Delta_{h,k}$ because they geometrically form a triangle when $h \leq k$, as shown in Figure 1.

Definition 1.2. Let $m = hk - h + 1$. Then define

$$\Delta_{h,k} \coloneqq \bigcup_{\ell=0}^{\min\{h,k\}-3} [m + \ell h + 1, m + \ell h + (h - 2 - \ell)].$$

Theorem 1.3. *For all $h, k \in \mathbb{N}$, we have*

$$\mathcal{R}(h,k) \cap \Delta_{h,k} = \emptyset.$$

Figure 1 shows an illustrated example of the points ruled out by Theorem 1.3. Tang–Xing's [TX21] theorem describes the intervals in $\Delta_{h,k}$ with $\ell = 0$ and $\ell = 1$ and $\ell = 2$, which are the uppermost three rows of the triangle in Figure 1, while Schinina's [Sch26] theorem describes the interval in $\Delta_{h,k}$ with $\ell = 0$, which is the uppermost row of the triangle in Figure 1. Our main theorem states that for general h and large k, all other numbers in the bounding interval outside $\Delta_{h,k}$ are in $\mathcal{R}(h,k)$.

Theorem 1.4. *Fix $h \in \mathbb{N}$. There exists a constant $k_h \in \mathbb{N}$ such that if $k > k_h$, then*

$$\mathcal{R}(h,k) = \left[hk - h + 1, \binom{h+k-1}{h}\right] \setminus \Delta_{h,k}.$$

Remark 1.5. Tracking constants in our proof shows that one can take $k_h \leq 2^{20h^2}$. We have not attempted to optimize this quantity further because it appears to be quite far from the truth.

Numerical experiments for small h and k suggest that perhaps the conclusion of Theorem 1.4 holds for all $k > h$.

Conjecture 1.6. Let $h \in \mathbb{N}$, and let $k > h$. Then

$$\mathcal{R}(h,k) = \left[hk - h + 1, \binom{h+k-1}{h}\right] \setminus \Delta_{h,k}.$$

When $k < h - 1$, the conclusion of Conjecture 1.6 never holds; this follows from results of Lev [Lev96]. See Section 5.2 for a discussion of what happens when $k \leq h$.

For any fixed h, Theorem 1.4 reduces Conjecture 1.6 to a finite problem, albeit a computationally intensive one. We are able to prove that $k_3 = 2$ suffices, which implies Conjecture 1.6 when $h = 3$; note that $\Delta_{3,k} = \{3k - 1\}$.

Theorem 1.7. *If $k > 2$, then*

$$\mathcal{R}(3,k) = \{3k-2\} \cup \left[3k, \binom{k+2}{3}\right].$$

In contrast with previous work, our proofs of Theorem 1.4 and Theorem 1.7 are nonconstructive; for a single desired sumset size, our proof does not give an explicit construction attaining it. Instead, we use continuity-type arguments to prove existence. We construct a sequence of sets A_i such that, even though the quantities $|hA_i|$ vary non-monotonically, we can guarantee that the entire set $\{|hA_i|\}$ is an interval. To study $|hA|$, we also study the distribution of the tuples $(|A|, |2A|, |3A|, \ldots, |hA|)$ for various k-element sets A. Even though we do not characterize exactly which tuples are possible, we extract enough structural information to characterize the possible values of $|hA|$. This opens up fruitful directions for new research, to control these tuples more precisely. Another key part of our construction is combining dense sets, meaning sets with small doubling, and sparse sets, which look like Sidon sets or B_h sets, to get sets in the middle.

Remark 1.8. We can also define $N(h,k)$ to be the minimal $N \geq 0$ such that

$$\mathcal{R}(h,k) = \{|hA| : A \subset [0,N] \text{ and } |A| = k\}.$$

By general considerations, Nathanson [Nat26b, Theorem 8] proved that $N(h,k) \leq 4(8h)^{k-1}$. Recently, ChatGPT 5.5 Pro, with light prompting from Timothy Gowers and the author, optimized the constructions in this paper to prove that $N(h,k)$ is polynomial in k. In a recent blog post [GR26], Gowers and the author describe the ChatGPT construction and intuition for why it works, detail how to show the explicit bound $N(h,k) \leq k^{10h^3}$ for k sufficiently large, and provide the fully ChatGPT-written proof that $N(h,k)$ is polynomial in k.

1.1 Roadmap

In Section 2, we prove Theorem 1.3 by using results from [Lev96] which generalize Freiman's $3k-3$ theorem to control the sizes of sumsets from the diameter of sets. We then prove Theorem 1.7 before Theorem 1.4 because the proof of Theorem 1.7 contains simpler versions of some of the ideas in the proof of Theorem 1.4. In Section 3, we prove Theorem 1.7 using a nonconstructive inductive argument involving understanding the possible tuples $(|2A|, |3A|)$ for all sets $A \subset \mathbb{Z}$ with $|A| = k$. The bulk of the paper is in Section 4, where we prove Theorem 1.4. This is done using two separate constructions: In Section 4.2, we construct sets A with smaller values of $|hA|$ using a partially inductive construction; in Section 4.3, we construct sets A with larger values of $|hA|$ by combining dense sets that resemble arithmetic progressions with sparse sets that resemble geometric sequences. Section 5 discusses possible future directions of research and open problems.

2 Ruling out Numbers from $\mathcal{R}(h,k)$

In this section, we will prove Theorem 1.3 quickly using results from [Lev96]. We begin by defining the diameter of a set A.

Definition 2.1. For $A \subset \mathbb{Z}$ finite, define its *diameter* $\operatorname{diam}(A)$ as 1 less than the minimum length of an arithmetic progression containing A.

Since affine linear transformations of A preserve both $|hA|$ and $\operatorname{diam}(A)$, we may always assume that $\min(A) = 0$ and $\max(A) = \operatorname{diam}(A)$ and $\gcd(A) = 1$. Write $A = \{0, a_1, \ldots, a_{k-1}\}$ with $a_i < a_{i+1}$ and $\gcd(a_1, \ldots, a_{k-1}) = 1$. Let $d \coloneqq a_{k-1} = \operatorname{diam}(A)$. Now, we state a result from [Lev96] about how $\operatorname{diam}(A)$ controls $|hA|$.

Theorem 2.2 ([Lev96])**.** *For any $A \subset \mathbb{Z}$ with $|A| = k \geq 2$ and $d = \operatorname{diam}(A)$ and $h \geq 2$,*

$$|hA| \geq |(h-1)A| + \min\{d, h(k-2)+1\}. \tag{2.1}$$

Let $q \in \mathbb{N}$ satisfy $q(k-2)+1 \leq d \leq (q+1)(k-2)+1$. By applying (2.1) inductively,

$$|hA| \geq |h\{0,1,2,\ldots,k-2,d\}| = \begin{cases} \frac{h(h+1)}{2}(k-2)+h+1 & \text{if } h \leq q \\ \frac{q(q+1)}{2}(k-2)+q+1+(h-q)d & \text{if } h \geq q. \end{cases} \tag{2.2}$$

When $h = 2$ and $q = 1$, (2.2) becomes Freiman's $3k-3$ theorem (see [Nat96, Theorem 1.13]). Using Theorem 2.2, we can prove Theorem 1.3.

Proof of Theorem 1.3. If $h \in \{1,2\}$ or $k \in \{1,2\}$ then $\Delta_{h,k} = \emptyset$, so let $h, k \geq 3$. Let A be a set of k integers, with $d = \operatorname{diam}(A)$. It suffices to show that $|hA| \notin \Delta_{h,k}$.

Case 1: $d < 2k-3$.

Let $\ell \in [0, \min\{h,k\}-3]$. Suppose $|hA| \geq m + \ell h + 1 = h(k-1+\ell)+2$. For cardinality reasons, $hA \not\subseteq [0, h(k-1+\ell)]$, so $d \geq k-\ell$. In Theorem 2.2, we have $q = 1$ so therefore (2.2) becomes

$$|hA| \geq k + (h-1)d \geq k + (h-1)(k-\ell) = m + \ell h + (h-\ell-1).$$

Therefore, $|hA|$ is not in the interval $[m+\ell h+1, m+\ell h+(h-\ell-2)]$. So $|hA| \notin \Delta_{h,k}$.

Case 2: $d \geq 2k-3$.

Theorem 2.2, specifically (2.1), implies that $|iA| - |(i-1)A| \geq 2k-3$ for all $i \geq 2$. Applying this $h-1$ times and using that $|A| = k$, then

$$|hA| \geq k + (h-1)(2k-3) = 2hk - 3h - k + 3. \tag{2.3}$$

As $\ell \leq k-3$ for all elements of $\Delta_{h,k}$,

$$\max(\Delta_{h,k}) \leq m + (k-3)h + (h-2-(k-3)) = 2hk - 3h - k + 2.$$

Therefore, $|hA| > \max(\Delta_{h,k})$ so $|hA| \notin \Delta_{h,k}$. □

3 Finding $\mathcal{R}(3,k)$

We begin by defining a useful graph, which we will use to formulate our inductive hypothesis.

Definition 3.1. Define the *graph representation* of a set $A \subset \mathbb{Z}$ as $g(A) \coloneqq (|2A|, |3A|)$. Define the *$k$-sumset graph* $\mathcal{G}_k$ to have vertex set

$$V_k = \{g(A) : A \subset \mathbb{Z} \text{ and } |A| = k\}.$$

Place an edge between distinct vertices (x,y) and (x',y') if $|x-x'| \leq 1$ and $|y-y'| \leq 1$.

As an example, the vertex set V_6 is shown in Figure 2. To prove Theorem 1.7, we will show inductively that for $k \geq 3$, the vertices $(2k, 3k)$ and $\left(\binom{k+1}{2}, \binom{k+2}{3}\right)$ are connected in $\mathcal{G}_k$; an example path is shown in Figure 2. Before starting the induction, we must prove that a particular part of the path between these vertices exists.

Lemma 3.2. *For $k \geq 4$, the points $(2k, 3k)$ and $(3k-2, 6k-5)$ are connected by a path in $\mathcal{G}_k$.*

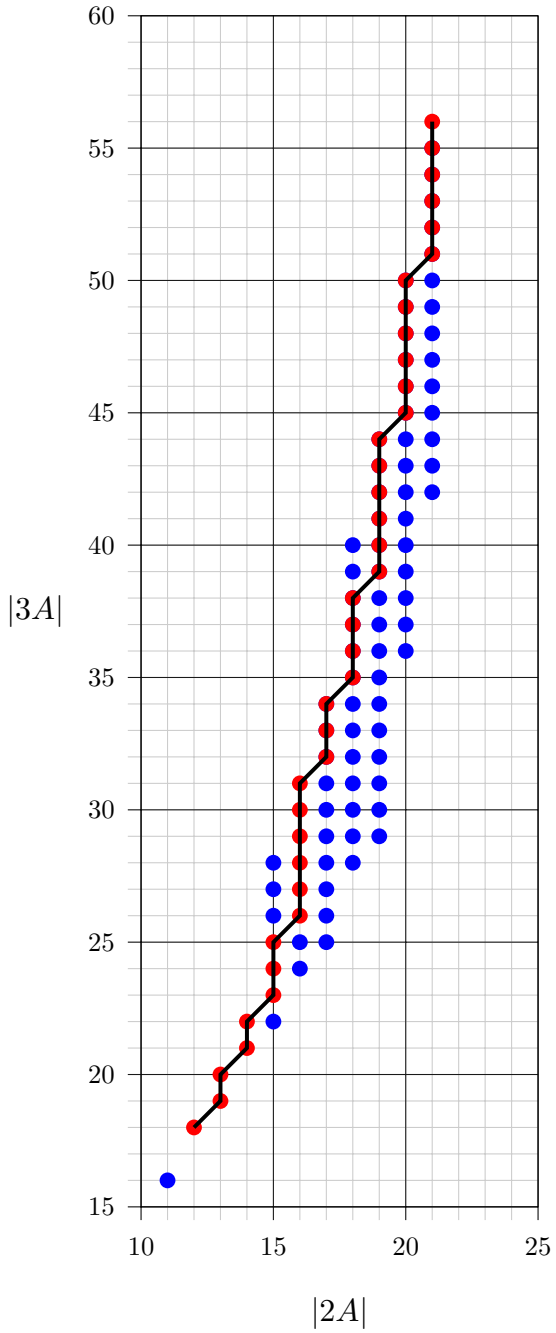


Figure 2: The blue and red points are all of the vertices of $\mathcal{G}_6$. These are the possible values of $g(A) = (|2A|, |3A|)$ when $|A| = 6$. The red points indicate the path constructed in the proof of Theorem 1.7 to connect $(12, 18)$ to $(21, 56)$.

Proof. We will explicitly construct a path $\mathcal{P}$ from $(2k, 3k)$ to $(3k-2, 6k-5)$. Define

$$A_a = \{0\} \cup [2, k-1] \cup \{k+a\},\ B_b = [0,1] \cup [3, k-1] \cup \{k+b\}.$$

Then explicit computations yield that

$$g(A_a) = (|2A_a|, |3A_a|) = \begin{cases} (2k+a, 3k+2a) & \text{if } 0 \le a \le k-3 \\ (3k-3, 5k-5) & \text{if } a = k-2 \\ (3k-2, 4k-2+a) & \text{if } k-1 \le a \le 2k-4 \\ (3k-2, 6k-5) & \text{if } a = 2k-2, \end{cases}$$

and

$$g(B_b) = (|2B_b|, |3B_b|) = \begin{cases} (2k+1+b, 3k+1+2b) & \text{if } 0 \le b \le k-4 \\ (3k-2, 5k-4) & \text{if } b = k-2, k > 4. \end{cases}$$

Also, define $B_2' = \{0,1,4,5\}$, and compute $g(B_2') = (9, 16)$. Let B_{k-2}^* denote B_{k-2} if $k > 4$ and B_2' if $k = 4$. Using the notation $g(C_1, \ldots, C_n) = g(C_1), \ldots, g(C_n)$, then we can check that

$$\mathcal{P} = g(A_0, B_0, A_1, B_1, \ldots, A_{k-4}, B_{k-4}, A_{k-3}, A_{k-2}, B_{k-2}^*, A_{k-1}, A_k, A_{k+1}, \ldots, A_{2k-5}, A_{2k-4}, A_{2k-2})$$

gives a path from $(2k, 3k)$ to $(3k-2, 6k-5)$ in $\mathcal{G}_k$, as desired. □

We are now ready to prove Theorem 1.7 by inducting on k, with inductive hypothesis the existence of a path between $(2k, 3k)$ and $\left(\binom{k+1}{2}, \binom{k+2}{3}\right)$ in $\mathcal{G}_k$.

Proof of Theorem 1.7. We induct on k. Our inductive hypothesis is that the vertices $(2k, 3k)$ and $\left(\binom{k+1}{2}, \binom{k+2}{3}\right)$ are connected by a path in $\mathcal{G}_k$. To prove Theorem 1.7 from this, it suffices to consider a path $\mathcal{P}$ from $(2k, 3k)$ to $\left(\binom{k+1}{2}, \binom{k+2}{3}\right)$. Looking at the y-values of the vertices in $\mathcal{P}$ implies that $\left[3k, \binom{k+2}{3}\right] \subseteq \mathcal{R}(3, k)$. Combining this with Theorem 1.3, which excludes $3k-1$ from $\mathcal{R}(3, k)$, proves Theorem 1.7.

The base case is $k = 3$, where it is easy to check that $g(\{0,1,3\})$ and $g(\{0,1,4\})$ form a path from $(6, 9)$ to $(6, 10)$ in $\mathcal{G}_3$. Now, let $k > 3$. By the inductive hypothesis, assume that $(2(k-1), 3(k-1))$ and $\left(\binom{k}{2}, \binom{k+1}{3}\right)$ are connected in $\mathcal{G}_{k-1}$. We will show that $(2k, 3k)$ and $\left(\binom{k+1}{2}, \binom{k+2}{3}\right)$ are connected in $\mathcal{G}_k$.

Let $A \subseteq [0, d]$ be a set with $|A| = k-1$ and $0, d \in A$. Define $A_1 = A \cup \{3d\}$ and $A_2 = A \cup \{3d+1\}$ to be sets with $|A_1| = |A_2| = k$. Then

$$3A_2 = 3A \cup (3d+1+2A) \cup (6d+2+A) \cup \{9d+3\}. \tag{3.1}$$

Since $hA \subseteq [0, hd]$ for $h \in \{1,2,3\}$, these four sets are disjoint and

$$|3A_2| = |3A| + |2A| + (k-1) + 1 = |3A| + |2A| + k.$$

With A_2 replaced by A_1, we have

$$3A_1 = 3A \cup (3d+2A) \cup (6d+A) \cup \{9d\}.$$

These four sets overlap in exactly one place, since $3d \in 3A \cap (3d+2A)$. So

$$|3A_1| = |3A| + |2A| + k - 1.$$

By similar logic, we can check that

$$|2A_1| = |2A_2| = |2A| + k.$$

For $i \in \{1,2\}$, we can define maps $\phi_i : V_{k-1} \to V_k$ by $\phi_i(g(A)) = g(A_i)$. Then

$$\phi_i((x,y)) = \left(x+k, x+y+k+\begin{cases} -1 & \text{if } i = 1 \\ 0 & \text{if } i = 2 \end{cases}\right).$$

For any set $S \subseteq V_{k-1}$, let

$$\phi(S) = \phi_1(S) \cup \phi_2(S).$$

Let $\phi(S) \subseteq \mathcal{G}_k$ refer to the induced subgraph of $\phi(S)$. Suppose (x,y) and (x',y') are adjacent in $\mathcal{G}_{k-1}$. By checking each value of $x-x'$ and $y-y'$ in $\{-1,0,1\}$, we can deduce that the graph $\phi(\{(x,y),(x',y')\}) \subseteq \mathcal{G}_k$ (consisting of up to four vertices) is connected.

Now, suppose (x, y) and (x', y') are connected in $\mathcal{G}_{k-1}$, so there is a path $\mathcal{P}$ from $(x, y) = (x_1, y_1)$ to $(x', y') = (x_n, y_n)$ with vertices $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$. So $\phi(\{(x_i, y_i), (x_{i+1}, y_{i+1})\}) \subseteq \mathcal{G}_k$ is connected for all $1 \leq i < n$. Therefore,

$$\phi(\{(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)\}) \subseteq \mathcal{G}_k$$

is connected, which means that $\phi_2((x, y))$ is connected to $\phi_2((x', y'))$ within $\mathcal{G}_k$.

Let $(x, y) = (2(k-1), 3(k-1))$ and $(x', y') = \left(\binom{k}{2}, \binom{k+1}{3}\right)$, which are connected in $\mathcal{G}_{k-1}$ by the inductive hypothesis. The previous paragraph demonstrates that $\phi_2((x, y)) = (3k-2, 6k-5)$ and $\phi_2((x', y')) = \left(\binom{k+1}{2}, \binom{k+2}{3}\right)$ are connected in $\mathcal{G}_k$. Lemma 3.2 gives that $(2k, 3k)$ and $(3k-2, 6k-5)$ are also connected in $\mathcal{G}_k$. By concatenating the paths from $(2k, 3k)$ to $(3k-2, 6k-5)$ and from $(3k-2, 6k-5)$ to $\left(\binom{k+1}{2}, \binom{k+2}{3}\right)$, we are done. □

4 Finding $\mathcal{R}(h, k)$ with General h

4.1 Overview

We introduce the statements of two main propositions, Proposition 4.1 and Proposition 4.2, which we will prove in Sections 4.2 and 4.3, respectively. Proposition 4.1 shows we can achieve the the smaller values in $\mathcal{R}(h, k)$, while Proposition 4.2 shows the same for the larger values.

Proposition 4.1. *Fix $h \in \mathbb{N}$, and let $k > 4^h h^2$. Then*

$$\left[hk - h + 1, \frac{k^h}{4^{2h^2}}\right] \setminus \Delta_{h,k} \subseteq \mathcal{R}(h, k).$$

Proposition 4.2. *Fix $h \geq 4$, and $\varepsilon \in \mathbb{R}_{>0}$. There exists some constant $k_{h,\varepsilon} \in \mathbb{N}$ such that for $k > k_{h,\varepsilon}$, we have*

$$\left[\varepsilon k^h, \binom{h+k-1}{h}\right] \subseteq \mathcal{R}(h, k).$$

Theorem 1.4 follows immediately from these propositions.

Proof of Theorem 1.4. With $h = 1$, the theorem is obvious. With $h = 2$, the theorem follows from [ES83]. With $h = 3$, the theorem follows from Theorem 1.7. Let $h \geq 4$. Let $\varepsilon = \frac{1}{4^{2h^2}}$. Setting $k_h = \max\{k_{h,\varepsilon}, 4^h h^2\}$ and letting $k > k_h$, we have

$$\left[hk - h + 1, \binom{h+k-1}{h}\right] \setminus \Delta_{h,k} \subseteq \mathcal{R}(h, k)$$

by Propositions 4.1 and 4.2. By Theorem 1.3, $\Delta_{h,k} \cap \mathcal{R}(h, k) = \emptyset$, so we are done. □

4.2 Achieving Small Sumset Sizes

In this section, we will prove Proposition 4.1. We begin by defining (h, d)-filling sets. Then, we will construct sets A as the union of an (h, d)-filling set of size $k - 1$ for some d and a single number. We will prove Proposition 4.1 by showing that varying d and the single number will cause hA to achieve all the desired sizes.

Definition 4.3. For $h \in \mathbb{N}$, a set $A \subseteq [0, d]$ is (h, d)-*filling* if $hA = [0, hd]$.

If A is (h, d)-filling, then $d \in A$, so it follows that $iA = [0, id]$ for all $i > h$. Therefore, A being (h, d)-filling implies that A is (i, d)-filling for all $i \geq h$. Also, if A is (h, d)-filling and $a \in [0, d]$, then $A \cup \{a\}$ is (h, d)-filling. We first show the existence of (h, d) filling sets of size k for all $d \ll \varepsilon k^h$, for some small fixed $\varepsilon > 0$.

Lemma 4.4. *Let $h \geq 2$ and $k \geq 4^{h+1}$ and*

$$d \in \left[k - 1, \frac{k^h}{4^{(h^2+h)/2}}\right].$$

Then there exists an (h, d)-filling set A with $|A| = k$ such that $[0, \frac{k}{8}] \subset A$.



Figure 3: An example of A' from the proof of Lemma 4.4 with $d = 100$.

Proof. Notice that $\frac{k^h}{4^{(h^2+h)/2}} \geq k$ since $k \geq 4^{h+1}$. We induct on h. The base case is $h = 2$.

With $h = 2$, then $d \in [k-1, \frac{k^2}{64}]$. Let $\tilde{d} = \lfloor \sqrt{d} \rfloor$. Consider the set

$$A' = [0, \tilde{d}] \cup \{i\tilde{d} : i \in \mathbb{Z} \text{ and } 0 \leq i \leq d/\tilde{d}\} \cup [d - \tilde{d}, d].$$

See Figure 3 for a picture of this construction. It is easy to check that A' is $(2, d)$-filling. Because $\tilde{d} \leq \frac{k}{8}$, we get $|A'| < \frac{k}{2}$. Now, create A from A' by adding the smallest $k - |A'|$ numbers in $[0, d] \setminus A'$ to A'. This is possible since $d \geq k - 1$. Then A remains $(2, d)$-filling. Since $k - |A'| > \frac{k}{2}$, we have that $[0, \frac{k}{8}] \subseteq [0, \frac{k}{2}] \subseteq A$.

Let $h > 2$ and assume that the lemma holds for $h - 1$. We will prove it for h. Let $k \geq 4^{h+1}$. If

$$d \leq \frac{k^{h-1}}{4^{(h^2-h)/2}},$$

then we can let A be the $(h-1, d)$-filling set constructed by induction, and we are done. So assume that

$$d \in \left[\frac{k^{h-1}}{4^{(h^2-h)/2}}, \frac{k^h}{4^{(h^2+h)/2}} \right].$$

Then

$$\left\lfloor \frac{4d}{k} \right\rfloor \in \left[\frac{k^{h-2}}{4^{(h^2-h)/2-1}} - 1, \frac{(k/4)^{h-1}}{4^{(h^2-h)/2}} \right].$$

Let $\tilde{d} = \lfloor \frac{4d}{k} \rfloor$ and let $\tilde{k} = \lceil \frac{k}{4} \rceil$, and notice $\tilde{d} \leq \tilde{k}^{h-1}/4^{(h^2-h)/2}$. If $\tilde{d} \geq \tilde{k} - 1$, the inductive hypothesis supplies an $(h-1, \tilde{d})$-filling set $\tilde{A}$ with $|\tilde{A}| = \tilde{k}$. If $\tilde{d} \leq \tilde{k} - 1$, let $\tilde{A} = [0, \tilde{d}]$ which is clearly $(h-1, \tilde{d})$-filling. Now, let

$$A' = \tilde{A} \cup \{i\tilde{d} : i \in \mathbb{Z} \text{ and } 0 \leq i \leq d/\tilde{d}\} \cup (d - \tilde{d} + \tilde{A}).$$

Each of the components of this union has size at most $\frac{k}{3.5}$, so $|A'| \leq \frac{7k}{8}$.

Let us now show that A' is (h, d)-filling. Since $\tilde{A}$ is $(h-1, \tilde{d})$-filling, we have

$$(h-1)\left(\tilde{A} \cup (d - \tilde{d} + \tilde{A})\right) = \bigcup_{j=0}^{h-1} \left(j(d - \tilde{d}) + [0, (h-1)\tilde{d}]\right).$$

It is then not hard to check that $hA' = [0, hd]$. To construct A, add the smallest $k - |A'|$ numbers between 0 and d to A', which is possible since $d \geq k - 1$. Then A remains (h, d)-filling. Because $k - |A'| \geq \frac{k}{8}$, we have $[0, \frac{k}{8}] \subseteq A$. □

We are now ready to prove Proposition 4.1 by taking A as the union of an (h, d)-filling set and a number.

Proof of Proposition 4.1. With $h = 1$ the proposition is vacuously true, so let $h \geq 2$. We will define two parameters d and i to vary, and we will show that varying these parameters gives all the desired sets.

Let $k - 2 \leq d \leq \frac{(k-1)^h}{4^{(h^2+h)/2}}$. Lemma 4.4 supplies a set B with $|B| = k - 1$ such that B is (h, d)-filling and $[0, \frac{k-1}{8}] \subseteq B$. Since $k > 4^h h^2$, we have $[0, h^2] \subseteq B$. Let $i \in [1, h]$, and define

$$A = \{0\} \cup (i + B).$$

Then, since $hi + hB = [hi, hi + hd]$, we have

$$hA = \{0\} \cup (i + B) \cup (2i + 2B) \cup (3i + 3B) \cup \cdots \cup [hi, hi + hd].$$

If $1 \leq m \leq h$, then $(mi + mB) \subseteq [mi, mi + md]$. Since $[0, h^2] \subseteq B$, we have $[i, hi] \subseteq (i + B)$, so

$$hA = \{0\} \cup [i, hi + hd].$$

Thus, $|hA| = h(i + d) - i + 2$. Define ℓ by $i + d = \ell + k$. So the conditions $d \in \left[k - 2, \frac{(k-1)^h}{4^{(h^2+h)/2}}\right]$ and $i \in [1, h]$ become

$$\ell \in \left[-1, \frac{(k-1)^h}{4^{(h^2+h)/2}} + h - k\right] \text{ and } i \in \left[\max\left\{1, \ell + k - \frac{(k-1)^h}{4^{(h^2+h)/2}}\right\}, \min\{h, \ell + 2\}\right].$$

Substituting $i+d$ for $\ell+k$ gives that $|hA| = hk+h\ell-i+2$. Then, recalling that $m = hk-h+1$, for any $\ell \in \left[-1, \frac{(k-1)^h}{4^{(h^2+h)/2}} - k\right]$, we have

$$\mathcal{R}(h,k) \supseteq hk+h\ell-[1,\min\{h,\ell+2\}]+2 = m+h\ell+[\max\{1,h-\ell-1\},h].$$

By plugging in the definition of $\Delta_{h,k}$, this gives that

$$\left[hk-h+1, h\left(\frac{(k-1)^h}{4^{(h^2+h)/2}} - k\right)\right] \setminus \Delta_{h,k} \subseteq \mathcal{R}(h,k),$$

and this immediately implies Proposition 4.1. □

4.3 Achieving Large Sumset Sizes

4.3.1 Overview

In this section, we will prove Proposition 4.2. We will first define a way to combine two sets A_1 and A_2 to get a set with $|A_1|+|A_2|$ elements, which we call the disjoint union $A_1 \sqcup A_2$. Our construction of sets A to fill the desired interval will be given as $A = B \sqcup C$, where $|B| = b$ and $|C| = c = k-b$, with $b \in [3, k-\lfloor k^{0.7} \rfloor]$. Here, B is a dense set ($|hB|$ is small for $h > 1$) and C is a sparse set ($|hC|$ is large for $h > 1$).

We define $B_{j,b} = [0, b-2] \cup \{h(b-2)+2-j\}$ with $j \in [1, (h-1)(b-2)+1]$ to be the union of an arithmetic progression and a single larger element. We also define the sets $S_{m,s_m} = \{0, 1, m, m^2, \ldots, m^{s_m-2}\}$ and $T_{m,t_m} = \{1, m, m^2, \ldots, m^{t_m-1}\}$ with powers of m, and then construct

$$C = \bigsqcup_{m=3}^{h} S_{m,s_m} \sqcup \bigsqcup_{m=2}^{h} T_{m,t_m}.$$

For $m \in [3,h]$, we vary $s_m \in [2, \lfloor c^{0.9} \rfloor]$ and $t_{m-1} \in [2, \lfloor c^{0.8} \rfloor]$, with $t_h = c - \sum_{m=3}^{h}(s_m + t_{m-1})$. Let

$$A = B_{j,b} \sqcup C = B_{j,b} \sqcup \bigsqcup_{m=3}^{h} S_{m,s_m} \sqcup \bigsqcup_{m=2}^{h} T_{m,t_m}. \tag{4.1}$$

We will show that varying the set of $2h-2$ parameters b, j, s_m, t_{m-1} for $m \in [3,h]$ forces $|hA|$ to achieve all the desired sizes.

We begin by stating a key auxiliary lemma.

Lemma 4.5. *Let $d \in \mathbb{N}$, and let $n_i \in \mathbb{N}$ for all $i \in [1,d]$. Let $f : [1,n_1] \times \cdots \times [1,n_d] \to \mathbb{Z}$ be a function which is nonincreasing in each entry. For $i \in [1,d]$, let*

$$\delta_i = \max(f(x_1,\ldots,x_{i-1},x_i,x_{i+1},\ldots,x_d) - f(x_1,\ldots,x_{i-1},x_i+1,x_{i+1},\ldots,x_d)) \geq 0$$

and

$$\Delta_i = \min(f(x_1,\ldots,x_{i-1},1,x_{i+1},\ldots,x_d) - f(x_1,\ldots,x_{i-1},n_i,x_{i+1},\ldots,x_d)) \geq 0.$$

Suppose that $\delta_1 = 1$ and $\delta_i \leq \Delta_{i-1}$ for all $i \in [2,d]$. Then $\mathrm{im}(f)$ is an interval.

To apply Lemma 4.5, let $d = 2h-3$, with

$$(x_1,x_2,x_3,\ldots,x_d) = (s_h-1, t_{h-1}-1, s_{h-1}-1, t_{h-2}-1, \ldots, s_3-1, t_2-1, j). \tag{4.2}$$

Fix $b \in [3, k-\lfloor k^{0.7} \rfloor]$ and $c = k-b$. Set

$$n_i = \begin{cases} \lfloor c^{0.9} \rfloor - 1 & \text{if } i \text{ is odd and } i < 2h-3 \\ \lfloor c^{0.8} \rfloor - 1 & \text{if } i \text{ is even} \\ (h-1)(b-2)+1 & \text{if } i = 2h-3. \end{cases} \tag{4.3}$$

Let $f(x_1,\ldots,x_d) = hA$, where A is defined as in (4.1). We will then show that hA is nonincreasing in each x_i, that $\delta_1 = 1$, and that $\Delta_{i-1} \geq \delta_i$ for $i \in [2,d]$. Let $\mathcal{A}_b$ to be the set of all A with b fixed and j, s_m, t_{m-1} for $m \in [3,h]$ varying in the prescribed intervals. By applying Lemma 4.5, the set $h\mathcal{A}_b \coloneqq \{|hA| : A \in \mathcal{A}_b\}$ is an interval.

We will then show that $h\mathcal{A}_b \cap h\mathcal{A}_{b-1} \neq \emptyset$, meaning that the intervals $h\mathcal{A}_b$ overlap for different values of b. Thus, with $\mathcal{A} = \bigcup_{b=3}^{k-\lfloor k^{0.7} \rfloor} \mathcal{A}_b$, the set $h\mathcal{A} \coloneqq \{|hA| : A \in \mathcal{A}\}$ is an interval. Finally, it suffices to show that

there is a value smaller than εk^h in $h\mathcal{A}_{k-\lfloor k^{0.7}\rfloor}$ and that $\binom{h+k-1}{h} \in h\mathcal{A}_3$. Since $h\mathcal{A}$ is an interval, it contains the whole interval we want, which finishes the proof.

In our construction, we can think of each x_i as a parameter to vary, where increasing each x_i adds more relations of sums of elements of A, decreasing $|hA|$. For smaller values of i, varying x_i gives finer control over the value of $|hA|$, while varying x_i for larger values of i gives coarser control over the value of $|hA|$.

We first prove Lemma 4.5. In Section 4.3.2, we define disjoint unions and a useful generating function, and show that the generating function behaves multiplicatively with respect to disjoint unions. We also define notions of $O[[\cdot]]$ and $\Theta_+[[\cdot]]$ for the size of generating functions. In Section 4.3.3, we approximate the sumset sizes of the components $B_{j,\ell}$, $S_{m,\ell}$, and $T_{m,\ell}$. In Section 4.3.4, we show that our construction satisfies the conditions of Lemma 4.5, and then use this to prove Proposition 4.2. Throughout this section, fix $h \geq 4$.

Proof of Lemma 4.5. We show that $f([1,n_1]\times\cdots\times[1,n_i]\times\{y_{i+1}\}\times\cdots\times\{y_d\})$ is an interval for any $i \in [1,d]$ and $y_{i+1} \in [1,n_{i+1}], \ldots, y_d \in [1,n_d]$ by induction on i. With $i=1$, this is true since $\delta_1 = 1$ and

$$f(a, y_2, \ldots, y_d) - \delta_1 \leq f(a+1, y_2, \ldots, y_d) \leq f(a, y_2, \ldots, y_d)$$

for any $a \in [1, n_1 - 1]$.

Now, assume the inductive hypothesis is true for $i-1$, and we will show it to be true for i. Define $I_{x_i} \coloneqq f([1,n_1]\times\cdots\times[1,n_{i-1}]\times\{x_i\}\times\{y_{i+1}\}\times\cdots\times\{y_d\})$ for $x_i \in [1,n_i]$, which is an interval by our inductive hypothesis. Then

$$f([1,n_1]\times\cdots\times[1,n_{i-1}]\times[1,n_i]\times\{y_{i+1}\}\times\cdots\times\{y_d\}) = \bigcup_{a=1}^{n_i} I_a. \tag{4.4}$$

Since f is nonincreasing in each entry, $\min I_a \geq \min I_{a+1}$ for all $a \in [1, n_i - 1]$. Also, by the definitions of δ_i and Δ_{i-1}, we have

$$f(1,1,1,\ldots,n_{i-1},a,y_{i+1},\ldots,y_d) \leq f(1,1,1,\ldots,1,a+1,y_{i+1},\ldots,y_d) + \delta_i - \Delta_{i-1}.$$

The value of f on the left side is in I_a and the value of f on the right side is in I_{a+1}. Since $\delta_i - \Delta_{i-1} \leq 0$, we have $\min I_a \leq \max I_{a+1}$. Hence, I_a and I_{a+1} overlap. So both sides of (4.4) are intervals, and we are done. □

4.3.2 Disjoint Unions and Generating Functions

First, we note that working in $\mathbb{Z}$ and $\mathbb{Z}^r$ are equivalent.

Lemma 4.6. *For $\eta, r, \kappa \in \mathbb{N}$, we have*

$$\{(|A|, |2A|, \ldots, |\eta A|) : A \subset \mathbb{Z}^r \text{ and } |A| = \kappa\} = \{(|A|, |2A|, \ldots, |\eta A|) : A \subset \mathbb{Z} \text{ and } |A| = \kappa\}.$$

Proof. We induct on r. With $r = 1$ this is trivial. Given $A \subset \mathbb{Z}^r$, let $M = \max\{\|a\|_\infty : a \in A\}$. Define a function $\phi : \mathbb{Z}^r \to \mathbb{Z}^{r-1}$ to be the linear map which takes $(x_1, \ldots, x_{r-1}, 0) \to (x_1, \ldots, x_{r-1})$ and takes $(0, 0, \ldots, 0, 1) \mapsto (\lceil 3\eta M \rceil, 0, 0, \ldots, 0)$. Let $A' = \phi(A)$. It is easy to check that ϕ is a Freiman isomorphism of order η, so we have $|iA'| = |iA|$ for $1 \leq i \leq \eta$. □

We now define a disjoint union of sets and a generating function.

Definition 4.7. Let $A \subset \mathbb{Z}^r$ and $B \subset \mathbb{Z}^s$ with $|A| = a$ and $|B| = b$. Define the *disjoint union* $A \sqcup B$ as the set of $a + b$ points in $\mathbb{Z}^{r+s+2}$ given by

$$A \sqcup B = (A, 0, 1, 0) \cup (0, B, 0, 1) \subset \mathbb{Z}^r \times \mathbb{Z}^s \times \mathbb{Z} \times \mathbb{Z}.$$

Definition 4.8. Define $0A = \{0\}$. For $A \subset \mathbb{Z}^r$ finite, define the generating function

$$\mathcal{F}_A(z) \coloneqq \sum_{\eta=0}^{\infty} |\eta A| z^\eta.$$

We now prove that the generating function behaves multiplicatively with respect to disjoint unions.

Lemma 4.9. *For $A \subset \mathbb{Z}^r$ and $B \subset \mathbb{Z}^s$ finite,*

$$\mathcal{F}_{A\sqcup B}(z) = \mathcal{F}_A(z) \cdot \mathcal{F}_B(z).$$

Proof. Elements of $\eta(A \sqcup B)$ are formed by adding up i elements from $(A, 0, 1, 0)$ and $\eta - i$ elements from $(0, B, 0, 1)$, which yields $|iA||(\eta - i)B|$ possible sums. Sums of η elements with different values of i cannot be equal, since they will differ in their last two coordinates. Varying the value of i between 0 and η gives

$$|\eta(A \sqcup B)| = |\eta A| + |(\eta - 1)A||B| + |(\eta - 2)A||2B| + \cdots + |\eta B|,$$

so we are done by applying Definition 4.8. □

Notice that $\mathcal{F}_{A\sqcup(B\sqcup C)}(z) = \mathcal{F}_{(A\sqcup B)\sqcup C}(z)$. Whenever taking a disjoint union of more than two sets, we do so in an arbitrary order, as in such circumstances we will only care about the data in the generating function, which we know to be equal. In particular, with $A = \bigsqcup_{i=1}^{m} A_i$, we have

$$\mathcal{F}_A(z) = \prod_{i=1}^{m} \mathcal{F}_{A_i}(z).$$

Recall that h is fixed; we now consider asymptotics as k grows to infinity. Given a function $f : \mathbb{N} \to \mathbb{Z}$ and $i \in \mathbb{R}_{\geq 0}$, we let the expressions $f(a) \in O(a^i)$ and $f(a) \in \Theta(a^i)$[1] have their usual meanings. Say that $f(a) \in \Theta_+(a^i)$ if $f(a) \in \Theta(a^i)$ and there is some $a_0 \in \mathbb{N}$ such that $f(a) > 0$ if $a > a_0$. Now, we define the notions of $O[[\cdot]]$ and $\Theta_+[[\cdot]]$ and $\equiv$ that we will use to approximate the sizes of generating functions.

Definition 4.10. We say that a power series $\mathcal{F}(z) = 1 + a_1 z + a_2 z^2 + a_3 z^3 + \cdots$ is in $O[[az]]$ (respectively $\Theta_+[[az]]$) if $a_i \in O(a^i)$ (respectively $\Theta_+(a^i)$) for $i \leq h$. For $\mathcal{G}(z) = 1 + b_1 z + b_2 z^2 + b_3 z^3 + \cdots$ another power series, we say $\mathcal{F}(z) \equiv \mathcal{G}(z)$ if $a_i = b_i$ for all $i \leq h$.

Note that for $\mathcal{F}(z) \in O[[az]]$ or $\Theta_+[[az]]$, we require $\mathcal{F}(z)$ to have constant term $a_0 = 1$. We now state a lemma about how the sizes of power series vary under multiplication.

Lemma 4.11. *Fix $r \in \mathbb{N}$. If $\mathcal{F}(z) \in \Theta_+[[az]]$ and $b = o(a)$ and $\mathcal{G}^{(j)}(z) \in O[[bz]]$ for all $j \in [1, r]$, then*

$$\mathcal{P}(z) \coloneqq \mathcal{F}(z) \cdot \prod_{j=1}^{r} \mathcal{G}^{(j)}(z) \in \Theta_+[[az]].$$

Proof. We induct on r. First let $r = 1$. Let $\mathcal{F}(z) = 1 + a_1 z + a_2 z^2 + a_3 z^3 + \cdots$ for $a_i \in \Theta_+(a^i)$. Let $\mathcal{G}^{(1)}(z) = 1 + b_1 z + b_2 z^2 + b_3 z^3 + \cdots$ for $b_i \in O(b^i)$. Let $\mathcal{P}(z) = 1 + p_1 z + p_2 z^2 + p_3 z^3 + \cdots$. For $\eta \leq h$, we have

$$p_\eta = \sum_{\nu=0}^{\eta} a_\nu b_{\eta-\nu}.$$

The fastest growing term will be $a_\eta \in \Theta_+(a^\eta)$. All other terms will be in $O(a^\eta) \cdot \frac{b}{a}$, and will thus be dwarfed by a_η. Thus, $p_\eta \in \Theta_+(a^\eta)$.

Now, assume the statement is true for $r - 1$. Let

$$\mathcal{P}'(z) \coloneqq \mathcal{F}(z) \cdot \prod_{j=1}^{r-1} \mathcal{G}^{(j)}(z) \in \Theta_+[[az]]$$

by the inductive hypothesis. By repeating the argument of $r = 1$ on $\mathcal{P}(z) = \mathcal{P}'(z)\mathcal{G}^{(r)}(z)$, we get that $\mathcal{P}(z) \in \Theta_+[[az]]$. □

4.3.3 Sumset Sizes of $B_{j,\ell}$, $S_{m,\ell}$, and $T_{m,\ell}$

We begin with our construction of dense sets $B_{j,\ell}$, which are given by the union of an arithmetic progression and an integer, and state their properties.

Definition 4.12. Let $\ell \geq 3$, and $s = (h-1)(\ell-2)+1$. Define the set $B_{j,\ell} \coloneqq [0, \ell-2] \cup \{h(\ell-2)+2-j\}$ for $j \in [1, s]$.

Lemma 4.13. *For $\ell \geq 3$, the sets $B_{1,\ell}, \ldots, B_{s,\ell}$ satisfy the following:*

(a) $0 \leq |\eta B_{j-1,\ell}| - |\eta B_{j,\ell}| \leq h$ for all $j \in [2, s]$ and $\eta \in [2, h]$;

[1] If $i < 0$ then $f(a) \in O(a^i)$ or $\Theta(a^i)$ if $f(a) = 0$ for all $a \in \mathbb{N}$.

(b) $|\eta B_{j,\ell}| < h^2\ell$ *for all* $j \in [1,s]$ *and* $\eta \in [2,h]$*;*

(c) $\mathcal{F}_{B_{1,3}}(z) \equiv (1-z)^{-3}$.

(d) If $\ell > 3$*, then* $\mathcal{F}_{B_{1,\ell}}(z) \equiv (1-z)^{-1}\mathcal{F}_{B_{s,\ell-1}}(z)$.

Proof. An exact formula for $|\eta B_{j,\ell}|$, due to [Lev96], is given in (2.2). This immediately implies (a). For (b), notice that $B_{j,\ell} \subseteq [0, h(\ell-1)]$ and $\eta \le h$, so $\eta B_{j,\ell} \subseteq [0, h^2(\ell-1)]$. Therefore $|\eta B_{j,\ell}| < h^2\ell$. For (c), the binomial theorem gives $(1-z)^{-3} = \sum_{\eta=0}^{\infty}\binom{\eta+2}{2}z^\eta$. By definition, $B_{1,3} = \{0,1,h+1\}$. For $1 \le \eta \le h$, the identity $|\eta B_{1,3}| = \binom{\eta+2}{2}$ follows from setting q to be h and h to be η and k to be 3 in (2.2).

For part (d), let $d = h(\ell-2)+1$. Then $B_{s,\ell-1} = [0,\ell-2]$ and $B_{1,\ell} = [0,\ell-2]\cup\{d\}$. Let $\eta \in [1,h]$. In a similar way to (3.1),

$$\eta B_{1,\ell} = \bigcup_{\nu=0}^{\eta}\big((\eta-\nu)d + \nu B_{s,\ell-1}\big). \tag{4.5}$$

Since $d > h(\ell-2)$, the sets on the right side of (4.5) are disjoint, so $|\eta B_{1,\ell}| = \sum_{\nu=0}^{\eta}|\nu B_{s,\ell-1}|$. Plugging in that $(1-z)^{-1} = (1+z+z^2+\cdots)$ yields (d). □

Now, we define a set $S_{m,\ell}$ to be a geometric series which includes 0, has common ratio m, and has ℓ elements. We then bound its generating function $\mathcal{F}_{S_{m,\ell}}(z)$.

Definition 4.14. For $m \ge 2$ and $\ell \ge 2$, define the set $S_{m,\ell} \coloneqq \{0,1,m,m^2,\ldots,m^{\ell-2}\}$.

Lemma 4.15. *For* $m \ge 2$ *and* $\ell \ge 2$*, we have*

$$\mathcal{F}_{S_{m,\ell}}(z) = \frac{1-(\ell-2)z^m O[[\ell z]]}{(1-z)^\ell}.$$

Proof. Let $\eta \in \mathbb{N}$. The elements in $\eta S_{m,\ell}$ can be viewed as numbers n where $\lfloor\frac{n}{m^{\ell-2}}\rfloor$ plus the sum of the base-m digits of $n \pmod{m^{\ell-2}}$ is less than or equal to η. To construct such an n, construct n with $\lfloor\frac{n}{m^{\ell-2}}\rfloor$ copies of $m^{\ell-2}$, and then use the base expansion of $n \pmod{m^{\ell-2}}$ in base m. This is a general construction of all elements n in $\eta S_{m,\ell}$.

This gives an expression of $|\eta S_{m,\ell}|$ as the number of ways to put η balls in ℓ boxes, where $\ell-2$ of the boxes have a maximum number of $m-1$ balls, and 2 of the boxes have no maximum. The $\ell-2$ boxes correspond to the number of 1's, m's, …, $m^{\ell-3}$'s in the base-m expansion of n. These numbers are bounded between 0 and $m-1$. One infinite-capacity box corresponds to the number of $m^{\ell-2}$'s in n, and the other corresponds to the 'overflow', or the number of 0's in n. In total, n must be formed by exactly η balls, as it is a sum of η elements of $S_{m,\ell}$. This gives a generating function formula

$$\mathcal{F}_{S_{m,\ell}}(z) = (1+z+z^2+\cdots+z^{m-1})^{\ell-2}(1+z+z^2+\cdots)^2 = \frac{(1-z^m)^{\ell-2}}{(1-z)^\ell}.$$

Expanding the numerator $(1-z^m)^{\ell-2}$ out using the binomial theorem yields the desired result. □

Now, we will define a similar set $T_{m,\ell}$, which is a geometric series with common ratio m and ℓ elements which does not include 0. With a bit more work, we prove a similar result about the generating function $\mathcal{F}_{T_{m,\ell}}(z)$.

Definition 4.16. For $m \ge 2$ and $\ell \ge 2$, define the set $T_{m,\ell} \coloneqq \{1,m,m^2,\ldots,m^{\ell-1}\} = S_{m,\ell+1}\setminus\{0\}$.

Lemma 4.17. *For* $m \ge 2$ *and* $\ell \ge 2$*, we have*

$$\mathcal{F}_{T_{m,\ell}}(z) = \frac{1-\binom{\ell-1}{2}z^{m+1}O[[\ell z]]}{(1-z)^\ell}.$$

Proof. Let $\eta \in \mathbb{N}$. We will approximate $|\eta T_{m,\ell}|$ by putting sums of elements of $T_{m,\ell}$ into a standardized form. Say that a representation $n = a_0 + a_1 m + a_2 m^2 + \cdots + a_{\ell-1}m^{\ell-1}$ is in *r-normal form* if $a_0,\ldots,a_{\ell-2}$ are in $[0,m-1]$ and $a_{\ell-1} \ge 0$ and $a_0+\cdots+a_{\ell-1} = r$. By taking the base-$m$ expansion of n, every $n \in \mathbb{N}$ can be written in normal form uniquely.

Each element $n \in \eta T_{m,\ell}$ can be written as $n = a_0 + a_1 m + a_2 m^2 + \cdots + a_{\ell-1}m^{\ell-1}$, where $a_i \ge 0$ and $a_0+\cdots+a_{\ell-1} = \eta$. This representation of n might not be unique, so we will put n into normal form. If $a_i \ge m$ for some $i < \ell-1$, we can decrement a_i by m and add 1 to a_{i+1} without changing the value of n; after this procedure then $a_0+\cdots+a_{\ell-1} = \eta-(m-1)$. Iterating this whenever possible, we will eventually get that $a_0,\ldots,a_{\ell-2}$ are in $[0,m-1]$ and $a_{\ell-1}$ is any nonnegative integer; if we have done this j times, we

get the sum $a_0+\cdots+a_{\ell-1}=\eta-j(m-1)$. Thus, any element $n\in\eta T_{m,\ell}$ can be put into $\eta-j(m-1)$-normal form for some $j\geq 0$, and this form is unique.

Now, we reverse this process. Consider an element n in $\eta-j(m-1)$-normal form. To pull back n into $\eta T_{m,\ell}$, we need to decrement a_i by 1 and increment a_{i-1} by m for j different nonzero values of i. This is possible as long as we never reach a state where $a_i=0$ for all $i>0$, which would mean n is represented as a sum of $\eta-j'(m-1)$ copies of 1. This will only happen when $n=\eta-j'(m-1)$ for $j'\geq 1$. This rules out $\lfloor\frac{\eta}{m-1}\rfloor$ numbers. Therefore, $|\eta T_{m,\ell}|$ is equal to the number of elements n in $\eta-j(m-1)$-normal form for any $j\geq 0$, minus $\lfloor\frac{\eta}{m-1}\rfloor$.

We now create the generating function $\mathcal{F}_{T_{m,\ell}}$. Elements in r-normal form can be counted as the number of ways to put r balls in ℓ boxes, where the number of balls in box i corresponds to a_i. There are $\ell-1$ boxes which have a maximum of $m-1$ balls, and 1 box with no maximum number of balls. Then, multiplying by $(1+z^{m-1}+z^{2(m-1)}+\cdots)$ to account for elements in $\eta-j(m-1)$-normal form, and subtracting $\lfloor\frac{\eta}{m-1}\rfloor$, gives the generating function:

$$\begin{aligned}\mathcal{F}_{T_{m,\ell}}(z)&=(1+z+z^2+\cdots+z^{m-1})^{\ell-1}(1+z+z^2+\cdots)(1+z^{m-1}+z^{2(m-1)}+\cdots)-\sum_{\eta=0}^{\infty}\left\lfloor\frac{\eta}{m-1}\right\rfloor z^\eta\\&=\frac{(1-z^m)^{\ell-1}}{(1-z)^\ell(1-z^{m-1})}-\sum_{\eta=0}^{\infty}\left\lfloor\frac{\eta}{m-1}\right\rfloor z^\eta.\end{aligned}$$

We can then simplify

$$\sum_{\eta=0}^{\infty}\left\lfloor\frac{\eta}{m-1}\right\rfloor z^\eta=z^{m-1}(1+z+z^2+\cdots)(1+z^{m-1}+z^{2(m-1)}+\cdots)=\frac{z^{m-1}}{(1-z)(1-z^{m-1})}.$$

Plugging this into our formula for $\mathcal{F}_{T_{m,\ell}}(z)$ gives

$$\mathcal{F}_{T_{m,\ell}}(z)=\frac{(1-z^m)^{\ell-1}-z^{m-1}(1-z)^{\ell-1}}{(1-z)^\ell(1-z^{m-1})}=\frac{1-z^{m-1}-\binom{\ell-1}{2}z^{m+1}+\sum_{i=3}^{\infty}O(\ell^i)z^{i+m-1}}{(1-z)^\ell(1-z^{m-1})}.$$

Expanding the left side of this out for $\ell=2$ and the right side out for $\ell>2$ yields the desired result. □

4.3.4 Putting it All Together

We begin by defining a notion of a family of sets being connected.

Definition 4.18. A set of sets $\mathcal{A}=\{A_1,\ldots,A_r\}$ with all $|A_i|=k$ is *connected* if the set of sumset sizes $h\mathcal{A}\coloneqq\{|hA|:A\in\mathcal{A}\}$ forms an interval.

Now, using all of our work in Section 4.3.3 to understand the component sets, we prove that the set of all A, with b fixed and j,s_m,t_{m-1} varying for $m\in[3,h]$, is connected.

Lemma 4.19. *Fix $h\geq 4$. Fix k to be sufficiently large, fix $b\in[3,k-\lfloor k^{0.7}\rfloor]$, and let $c=k-b$. Let $B_j\coloneqq B_{j,b}$. Define the family of sets*

$$\mathcal{A}_b=\left\{A=B_j\sqcup\bigsqcup_{m=3}^{h}S_{m,s_m}\sqcup\bigsqcup_{m=2}^{h}T_{m,t_m}:\begin{smallmatrix}j\in[1,(h-1)(b-2)+1]\\ s_m\in[2,\lfloor c^{0.9}\rfloor]\text{ if }m\in[3,h]\\ t_{m-1}\in[2,\lfloor c^{0.8}\rfloor]\text{ if }m\in[3,h]\\ t_h=c-\sum_{m=3}^{h}(s_m+t_{m-1})\end{smallmatrix}\right\}.$$

Then $\mathcal{A}_b$ is connected.

Proof. We will apply Lemma 4.5 with $d=2h-3$ and

$$(x_1,x_2,x_3,\ldots,x_d)=(s_h-1,t_{h-1}-1,s_{h-1}-1,t_{h-2}-1,\ldots,s_3-1,t_2-1,j).$$

The values n_i are given in (4.3). So it will suffice to show that $|hA|$ is nonincreasing in each entry x_i, that $\delta_1=1$, and that $\delta_i\leq\Delta_{i-1}$ for $i\in[2,d]$.

Let $C=\bigsqcup_{m=3}^{h}S_{m,s_m}\sqcup\bigsqcup_{m=2}^{h}T_{m,t_m}$. By Lemmas 4.15 and 4.17, we have that

$$\mathcal{F}_C(z)=\frac{1}{(1-z)^c}\prod_{m=3}^{h}(1-(s_m-2)z^mO[[s_mz]])\cdot\prod_{m=2}^{h}\left(1-\binom{t_m-1}{2}z^{m+1}O[[t_mz]]\right).\tag{4.6}$$

By the binomial theorem,

$$\frac{1}{(1-z)^c} = \sum_{\eta=0}^{\infty} \binom{\eta+c-1}{\eta} z^\eta \in \Theta_+[[cz]]. \tag{4.7}$$

We now prove the following two assertions about how changing one s_μ or t_μ while keeping all of the other s_m and t_{m-1} unchanged for $m \in [3, h-1]$ will affect $\mathcal{F}_C(z)$.

($\star$) For $\mu \in [3, h]$, increasing s_μ by 1 decreases $\mathcal{F}_C(z)$ by an element of $z^\mu \Theta_+[[cz]]$.

($\star\star$) For $\mu \in [2, h-1]$, increasing t_μ by 1 decreases $\mathcal{F}_C(z)$ by an element of $(t_\mu - 1)z^{\mu+1}\Theta_+[[cz]]$.

Notice that we can ignore the factor coming from t_h in (4.6) since it only affects powers of z greater than or equal to $h+1$, and also notice that each of the other factors in the products on the right side of (4.6) is in $O[[c^{0.9}z]]$. To prove ($\star$) and ($\star\star$), we now use Lemma 4.11 to combine all of the factors in the products on the right side of (4.6). For ($\star$), increasing s_μ by 1 to $s'_\mu = s_\mu + 1$ will decrease $\mathcal{F}_C(z)$ by an element of

$$\Theta_+[[cz]]((s'_\mu - 2) - (s_\mu - 2))z^\mu O[[s_\mu z]] \prod_{m \in [3,h]\setminus\{\mu\}} O[[c^{0.9}z]] \cdot \prod_{m=2}^{h-1} O[[c^{0.9}z]] = z^\mu \Theta_+[[cz]].$$

For ($\star\star$), increasing t_μ by 1 to $t'_\mu = t_\mu + 1$ decreases $\mathcal{F}_C(z)$ by an element of

$$\Theta_+[[cz]] \left(\binom{t'_\mu - 1}{2} - \binom{t_\mu - 1}{2} \right) z^{\mu+1} O[[t_\mu z]] \prod_{m=3}^{h} O[[c^{0.9}z]] \cdot \prod_{m \in [2,h-1]\setminus\{\mu\}} O[[c^{0.9}z]] = (t_\mu - 1)z^{\mu+1}\Theta_+[[cz]].$$

Since $A = B_j \sqcup C$, we have

$$|hA| = \sum_{\eta=0}^{h} |\eta C||(h-\eta)B_j|. \tag{4.8}$$

By ($\star$) and ($\star\star$) and Lemma 4.13(a), we have that $|hA|$ is nonincreasing in each entry x_i. From ($\star$) and (4.8), it is easy to check that the only term affected by increasing s_h by 1 will be $|hC|$, which will decrease by 1. Hence, $\delta_1 = 1$.

We now approximate δ_i for $i \in [1, d]$ and Δ_i for $i \in [1, d-1]$, which we will use to prove that $\delta_i \le \Delta_{i-1}$ for $i \in [2, d]$, which implies the lemma. Throughout, we use Lemma 4.13(b) to show that the terms with $\eta = h$ and $\eta = h-1$ dominate in (4.8).

Case 1: i is odd and $i < d$. Let $x_i = s_\mu - 1$ for $\mu \in [3, h]$. By ($\star$), increasing s_μ by 1 will decrease $\mathcal{F}_C(z)$ by an element of $z^\mu \Theta_+[[cz]]$. By (4.8),

$$\delta_i = \sum_{\eta=\mu}^{h} \Theta_+(c^{\eta-\mu})|(h-\eta)B_j| = \Theta_+(c^{h-\mu}) + b\Theta_+(c^{h-\mu-1}). \tag{4.9}$$

So, increasing s_μ from 2 to $\lfloor c^{0.9} \rfloor$ gives that

$$\Delta_i = \left(\lfloor c^{0.9} \rfloor - 2\right) \sum_{\eta=\mu}^{h} \Theta_+(c^{\eta-\mu})|(h-\eta)B_j| = \Theta_+(c^{h-\mu+0.9}) + b\Theta_+(c^{h-\mu-0.1}). \tag{4.10}$$

Case 2: i is even. Let $x_i = t_\mu - 1$ for $\mu \in [2, h-1]$. By ($\star\star$), increasing t_μ by 1 will decrease $\mathcal{F}_C(z)$ by an element of $(t_\mu - 1)z^{\mu+1}\Theta_+[[cz]]$. By (4.8),

$$\delta_i = \left(\lfloor c^{0.8} \rfloor - 1\right) \sum_{\eta=\mu+1}^{h} \Theta_+(c^{\eta-\mu-1})|(h-\eta)B_j| = \Theta_+(c^{h-\mu-0.2}) + b\Theta_+(c^{h-\mu-1.2}). \tag{4.11}$$

By ($\star\star$), increasing t_μ from 2 to $\lfloor c^{0.8} \rfloor$ will decrease $\mathcal{F}_C(z)$ by an element of $\Theta_+(c^{1.6})z^{\mu+1}\Theta_+[[cz]]$. By (4.8),

$$\Delta_i = \sum_{\eta=\mu+1}^{h} \Theta_+(c^{\eta-\mu-1+1.6})|(h-\eta)B_j| \ge \Theta_+(c^{h-\mu+0.6}). \tag{4.12}$$

Case 3: $i = d$. Here, $x_d = j$. By (4.8) and Lemma 4.13(a),

$$\delta_d = \max_{j \in [1,(h-1)(b-2)]} \sum_{\eta=0}^{h-2} |\eta C|\left(|(h-\eta)B_{j+1}| - |(h-\eta)B_j|\right) \le h \sum_{\eta=0}^{h-2} |\eta C| \le \Theta_+(c^{h-2}). \tag{4.13}$$

Now, we can compare the formulas of (4.9), (4.10), (4.11), (4.12), and (4.13) to show $\delta_i \le \Delta_{i-1}$ for $i \in [2,d]$. With i being odd and less than d, let $x_i = s_\nu - 1$ and $x_{i-1} = t_\nu - 1$. Then we are comparing (4.9) with $\mu = \nu$ to (4.12) with $\mu = \nu$. Since $3 \le b \le k - \lfloor k^{0.7} \rfloor$, and $c = k - b$, we have $c^{1.5} > b$. Hence, $\delta_i \le \Delta_{i-1}$. With i being even, let $x_i = t_\nu - 1$ and $x_{i-1} = s_{\nu+1} - 1$. Then we are comparing (4.11) with $\mu = \nu$ to (4.10) with $\mu = \nu + 1$, in which case it is clear that $\delta_i \le \Delta_{i-1}$. With $i = d$, we are comparing (4.13) to (4.12) with $\mu = 2$, so it is clear that $\delta_d \le \Delta_{d-1}$.

So, we have checked all of the assumptions of Lemma 4.5. Applying Lemma 4.5 yields the result. □

We are now ready to prove Proposition 4.2 by showing that the values of $|hA|$ for $A \in \mathcal{A}_b$ with b varying form an interval, and this interval contains a value smaller than εk^h and also contains $\binom{h+k-1}{h}$.

Proof of Proposition 4.2. Let $\mathcal{A}_b$ be as in Lemma 4.19 and recall that $c = k - b$. Define

$$\mathcal{A} \coloneqq \bigcup_{b=3}^{k-\lfloor k^{0.7} \rfloor} \mathcal{A}_b.$$

By Lemma 4.19, $\mathcal{A}_b$ is connected. To show that $\mathcal{A}$ is connected, it suffices to show that $h\mathcal{A}_b \cap h\mathcal{A}_{b-1} \neq \emptyset$ for $b \in [4, k - \lfloor k^{0.7} \rfloor]$.

For any $b \in [3, k - \lfloor k^{0.7} \rfloor]$, define a representative set $A_b \in \mathcal{A}_b$ by taking $(x_1, \dots, x_d) = (1, \dots, 1, 1)$ using (4.1) and (4.2). For any $b \in [4, k - \lfloor k^{0.7} \rfloor]$, define another representative set $A'_{b-1} \in \mathcal{A}_{b-1}$ by taking $(x_1, \dots, x_d) = (1, \dots, 1, n_d)$ using (4.1) and (4.2). Let $A_b = B_{1,b} \sqcup C$ and $A'_{b-1} = B_{s,b-1} \sqcup C'$. As $s_\mu = t_{\mu-1} = 2$ for all $\mu \in [3, h]$ for C, (4.6) gives that $\mathcal{F}_C(z) \equiv (1-z)^{-|C|} = (1-z)^{-(k-b)}$. Similarly, $\mathcal{F}_{C'}(z) \equiv (1-z)^{-|C'|} = (1-z)^{-(k-b+1)}$. By Lemma 4.13(d), we have $\mathcal{F}_{B_{1,b}}(z) \equiv (1-z)^{-1} \mathcal{F}_{B_{s,b-1}}(z)$. Hence,

$$\mathcal{F}_{A_b}(z) = \mathcal{F}_{B_{1,b}}(z)\mathcal{F}_C(z) \equiv \mathcal{F}_{B_{s,b-1}}(z)(1-z)^{-1}(1-z)^{-(k-b)} \equiv \mathcal{F}_{B_{s,b-1}}(z)\mathcal{F}_{C'}(z) = \mathcal{F}_{A'_{b-1}}(z),$$

so $|hA_b| = |hA'_{b-1}|$. So $h\mathcal{A}_b \cap h\mathcal{A}_{b-1} \neq \emptyset$.

We now show that $\binom{h+k-1}{h} \in h\mathcal{A}_3$ and that there is a value smaller than εk^h in $h\mathcal{A}_{k-\lfloor k^{0.7} \rfloor}$. This will imply that the interval $h\mathcal{A}$ contains the desired interval. Let $A_3 = B_{1,3} \sqcup C$, so by similar logic to the previous paragraph we get $\mathcal{F}_C(z) \equiv (1-z)^{-(k-3)}$. Plugging into Lemma 4.13(c) gives the formula

$$\mathcal{F}_{A_3}(z) \equiv \mathcal{F}_{B_{1,3}}(z)\mathcal{F}_C(z) \equiv (1-z)^{-3}(1-z)^{-(k-3)} \equiv (1-z)^{-k}.$$

By the binomial theorem, $|hA_3| = \binom{h+k-1}{h} \in h\mathcal{A}_3$.

In a similar fashion, we get that

$$\mathcal{F}_{A_{k-\lfloor k^{0.7} \rfloor}}(z) \equiv (1-z)^{-\lfloor k^{0.7} \rfloor} \mathcal{F}_{B_{1,k-\lfloor k^{0.7} \rfloor}}(z). \tag{4.14}$$

By Lemma 4.13(b), $|\eta B_{1,k-\lfloor k^{0.7} \rfloor}| \in O(k)$ for all $1 \le \eta \le h$. By (4.7), $(1-z)^{-\lfloor k^{0.7} \rfloor} \in \Theta_+[[k^{0.7}z]]$. Multiplying out the z^h term of (4.14) gives that $|hA_{k-\lfloor k^{0.7} \rfloor}| \in O(k^{0.7(h-1)+1})$. Since $0.7(h-1)+1 < h$, we can get $|hA_{k-\lfloor k^{0.7} \rfloor}| < \varepsilon k^h$ for any desired $\varepsilon > 0$ by making k large enough. □

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References

[ES83] Paul Erdős and Endre Szemerédi. On sums and products of integers. In *Studies in pure mathematics*, pages 213–218. Birkhäuser, Basel, 1983.

[GR26] Timothy Gowers and Isaac Rajagopal. A recent experience with ChatGPT 5.5 Pro. *Gowers's Weblog*, 2026. https://gowers.wordpress.com/2026/05/08/a-recent-experience-with-chatgpt-5-5-pro/.

[Lev96] Vsevolod F. Lev. Structure theorem for multiple addition and the Frobenius problem. *J. Number Theory*, 58(1):79–88, 1996.

[Nat96] Melvyn B. Nathanson. *Additive number theory: Inverse problems and the geometry of sumsets*, volume 165 of *Graduate Texts in Mathematics*. Springer-Verlag, New York, 1996.

[Nat25a] Melvyn B. Nathanson. Additive sumset sizes with tetrahedral differences, 2025. arXiv: 2507.08646.

[Nat25b] Melvyn B. Nathanson. Explicit sumset sizes in additive number theory. *Canad. Math. Bull.*, 176:1–12, 2025.

[Nat25c] Melvyn B. Nathanson. Problems in additive number theory, VI: Sizes of sumsets. *Acta Math. Hungar.*, 176:498–521, 2025.

[Nat26a] Melvyn B. Nathanson. Triangular and tetrahedral number differences of sumset sizes in additive number theory. *Integers*, 26, 2026.

[Nat26b] Melvyn B. Nathanson. Compression and complexity for sumset sizes in additive number theory. *J. Number Theory*, 281:321–343, 2026.

[O'B25] Kevin O'Bryant. On Nathanson's triangular number phenomenon, 2025. arXiv: 2506.20836.

[Plü70] Helmut Plünnecke. Eine zahlentheoretische anwendung der graphentheorie. *J. Reine Angew. Math.*, 243:171–183, 1970.

[Raz08] Alexander A. Razborov. On the minimal density of triangles in graphs. *Combin. Probab. Comput.*, 17(4):603–618, 2008.

[Ruz89] Imre Z. Ruzsa. An application of graph theory to additive number theory. *Sci. Ser. A Math. Sci. (N.S.)*, 3:97–109, 1989.

[Sch26] Vincent Schinina. On the sumset of sets of size k. *Integers*, 26, 2026.

[TX21] Min Tang and Yun Xing. Some inverse results of sumsets. *Bull. Korean Math. Soc.*, 58(2):305–313, 2021.

[Zha23] Yufei Zhao. *Graph theory and additive combinatorics—exploring structure and randomness*. Cambridge University Press, Cambridge, 2023.